

PRAKAMYA MISHRA

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EDUCATION

University of Massachusetts Amherst

Masters of Science in Computer Science (Data Science Concentration)

Sep. 2021 - May 2023

Amherst, MA, USA

Shiv Nadar University

B.Tech. in Computer Science & Engineering

Sep. 2016 - May 2020

Noida, UP, India

WORK EXPERIENCE

AMD, GenAI Team

Jan. 2024 – Present

Applied Research Engineer

Seattle, WA, USA

- Working on training, fine-tuning, and aligning large language models (LLMs) by exploring novel architectures and large-scale training techniques to reach state-of-the-art performance.
- Designing and implementing novel research ideas and improving LMMs to influence future hardware and software direction at AMD AI.
- Core contributor for **AMD-OLMo-1B**, **Instella-3B**, **Instella-3B-Long**, and **Instella-3B-Math**.

UMass BioNLP Lab

Jan. 2023 – Dec. 2024

Research Assistant

Amherst, MA, USA

- Proposed novel RLHF approaches that utilizes LLMs to generate high-quality feedback aimed at enhancing factual consistency in clinical note summarization.

Amazon, Alexa NLU Team

Jan. 2023 – Jul. 2023

Graduate Student Researcher

Amherst, MA, USA

- Proposed a novel clustering-based few-shot sampling approach for improving the performance of the present SOTA keyphrase extraction model in cross-domain setting.

Amazon, Alexa SLU Team

May 2022 – Aug. 2022

Applied Scientist Intern

Boston, MA, USA

- Developed a novel multi-cross attention transformer model for incorporating multi-modal input signals from multiple transformer encoders to improve Alexa SLU pipeline performance by x7 times.

SELECTED PUBLICATIONS

- SAND-Math: Using LLMs to Generate Novel, Difficult and Useful Mathematics Questions and Answers**
Chaitanya Manem, Pratik Prabhanjan Brahma, **Prakamya Mishra**, Zicheng Liu, Emad Barsoum
Preprint, arXiv, 2025
- TTT-Bench: A Benchmark for Evaluating Reasoning Ability with Simple and Novel Tic-Tac-Toe-style Games**
Prakamya Mishra*, Jiang Liu, Jialian Wu, Xiaodong Yu, Zicheng Liu, Emad Barsoum
EMNLP 2025, Main
- SYNFAC-EDIT: Synthetic Imitation Edit Feedback for Factual Alignment in Clinical Summarization**
Prakamya Mishra*, Zonghai Yao*, Parth Vashisht, Feiyan Ouyang, Beining Wang, Vidhi Dhaval Mody, Hong Yu
EMNLP 2024, Main
- Clustering-based Sampling for Few-Shot Cross-Domain Keyphrase Extraction**
Prakamya Mishra*, Lincy Pattanaik*, Arunima Sundar*, Nishant Yadav, Mayank Kulkarni
EACL 2024, Findings
- Synthetic Imitation Edit Feedback for Factual Alignment in Clinical Summarization**
Prakamya Mishra*, Zonghai Yao*, Shuwei Chen, Beining Wang, Rohan Mittal, Hong Yu
Workshop on SyntheticData4ML @ NeurIPS 2023
- NeuralNERE: Neural Named Entity Relationship Extraction for End-to-End Climate Change Knowledge Graph Construction**
Prakamya Mishra, Rohan Mittal
Workshop on Tackling Climate Change with Machine Learning @ ICML 2021

- **Bi-ISCA: Bidirectional Inter-Sentence Contextual Attention Mechanism for Detecting Sarcasm in User Generated Noisy Short Text**
Prakamya Mishra, Saroj Kaushik, Kuntal Dey
Workshop on Modelling and Reasoning in Context @ IJCAI 2021
- **STEPS-RL: Speech-Text Entanglement for Phonetically Sound Representation Learning**
Prakamya Mishra
PAKDD 2021

SKILLS

AI Research: Python, Pytorch, Transformers, Numpy, Pandas, Matplotlib, Shell, Pre/Post-Training, Evaluation